

Собираем свои модули

Для начала нам нужно создать файл **test1.c**, в который пропишем код нашего простого модуля. Файл будет доступен в той же папке репозитория, где лежит отчет.

Пишем также **Makefile**, который будет собирать наш модуль. Его прикладываю также в репозитории.

Запускаем make:

```
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ make
make -C /lib/modules/6.8.0-124-generic/build M=/home/vlad/eltex/hw_20 modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-124-generic'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using: gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] /home/vlad/eltex/hw_20/test1.o
MODPOST /home/vlad/eltex/hw_20/Module.symvers
CC [M] /home/vlad/eltex/hw_20/test1.mod.o
LD [M] /home/vlad/eltex/hw_20/test1.ko
BTF [M] /home/vlad/eltex/hw_20/test1.ko
Skipping BTF generation for /home/vlad/eltex/hw_20/test1.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-124-generic'
```

Посмотрим логи с помощью команды **dmesg | tail -n10**:

```
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ dmesg | tail -n10
[ 3.366053] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.367052] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.368393] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 4.236521] systemd-journald[40]: Collecting audit messages is disabled.
[ 4.282013] systemd-journald[40]: Received client request to flush runtime journal.
[ 4.286939] systemd-journald[40]: File /var/log/journal/6b2382371e6948568ec29ea7043c9db6/system.journal corrupted or
uncleanly shut down, renaming and replacing.
[ 5.787080] systemd-journald[40]: /var/log/journal/6b2382371e6948568ec29ea7043c9db6/user-1000.journal: Journal file u
ses a different sequence number ID, rotating.
[ 6.246211] WSL (173) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 48.349479] hv_balloon: Max. dynamic memory size: 8044 MB
[ 48.648226] systemd-journald[40]: Time jumped backwards, rotating.
vlad@DESKTOP-MKU600H:~/eltex/hw_20$
```

Всё пусто. Теперь зарегистрируем наш модуль командой **sudo insmod test1.ko** и проверим логи:

```
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ dmesg | tail -5
[ 3.958427] systemd-journald[44]: File /var/log/journal/6b2382371e6948568ec29ea7043c9db6/system.journal corrupted or
uncleanly shut down, renaming and replacing.
[ 5.599839] systemd-journald[44]: /var/log/journal/6b2382371e6948568ec29ea7043c9db6/user-1000.journal: Journal file u
ses a different sequence number ID, rotating.
[ 6.685824] WSL (171) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 28.453553] test1: loading out-of-tree module taints kernel.
[ 28.455131] My test module loaded!!!!
```

Как видим, мой тест модуль загрузился. Теперь удалим его и вновь посмотрим логи.

\$ sudo rmmod test1

\$ dmesg | tail -5

```
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ sudo rmmod test1
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ dmesg | tail -5
[ 6.685824] WSL (171) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 28.453553] test1: loading out-of-tree module taints kernel.
[ 28.455131] My test module loaded!!!!
[ 49.087795] hv_balloon: Max. dynamic memory size: 8044 MB
[ 154.665525] My test module unloaded!!!
vlad@DESKTOP-MKU600H:~/eltex/hw_20$
```

Готово, модуль выгрузился.

Приступаем ко второй части задания. Для этого создадим **test2.c**, пропишем **make, sudo insmod test2.ko** и выведем логи **dmesg | tail -n10**:

```

vlad@DESKTOP-MKU600H:~/eltex/hw_20$ dmesg | tail -n10
[ 3.952862] systemd-journald[44]: Received client request to flush runtime journal.
[ 3.958427] systemd-journald[44]: File /var/log/journal/6b2382371e6948568ec29ea7043c9db6/system.journal corrupted or
uncleanly shut down, renaming and replacing.
[ 5.599839] systemd-journald[44]: /var/log/journal/6b2382371e6948568ec29ea7043c9db6/user-1000.journal: Journal file u
ses a different sequence number ID, rotating.
[ 6.685824] WSL (171) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 28.453553] test1: loading out-of-tree module taints kernel.
[ 28.455131] My test module loaded!!!!
[ 49.087795] hv_balloon: Max. dynamic memory size: 8044 MB
[ 154.665525] My test module unloaded!!!
[ 1658.318374] My test module loaded!!!!
[ 1658.318382] Major number is 239
vlad@DESKTOP-MKU600H:~/eltex/hw_20$

```

Теперь создадим специальный файл символического устройства:

```
$ sudo mknod test2 c 239 0
```

И проверим созданся ли он командой `ls -la test2`

```

vlad@DESKTOP-MKU600H:/dev$ sudo mknod test2 c 239 0

vlad@DESKTOP-MKU600H:/dev$ ls -la test2
crw-r--r-- 1 root root 239, 0 Jun 16 16:13 test2
vlad@DESKTOP-MKU600H:/dev$

```

Изменим через него нашу строку:

```

vlad@DESKTOP-MKU600H:/dev$ cat test2
cat: test2: Bad address
vlad@DESKTOP-MKU600H:/dev$ head test2
Hello
vlad@DESKTOP-MKU600H:/dev$ sudo echo "WWWWWWAAAASS" > test2
bash: test2: Permission denied
vlad@DESKTOP-MKU600H:/dev$ sudo chmod +666 test2
vlad@DESKTOP-MKU600H:/dev$ sudo echo "WWWWWWAAAASS" > test2
vlad@DESKTOP-MKU600H:/dev$ head test2
WWWWWWAAAASS
vlad@DESKTOP-MKU600H:/dev$

```

Все работает. Теперь удалим созданное устройство:

```
$ sudo rmmod test2
```

```

vlad@DESKTOP-MKU600H:/dev$ sudo rmmod test2
vlad@DESKTOP-MKU600H:/dev$ dmesg | tail -n10
[ 5.599839] systemd-journald[44]: /var/log/journal/6b2382371e6948568ec29ea7043c9db6/user-1000.journal: Journal file u
ses a different sequence number ID, rotating.
[ 6.685824] WSL (171) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 28.453553] test1: loading out-of-tree module taints kernel.
[ 28.455131] My test module loaded!!!!
[ 49.087795] hv_balloon: Max. dynamic memory size: 8044 MB
[ 154.665525] My test module unloaded!!!
[ 1658.318374] My test module loaded!!!!
[ 1658.318382] Major number is 239
[ 1742.970512] WSL (181) ERROR: No buffer space available @telemetry.cpp:190 (StartTelemetryAgent)
[ 2062.436659] My test module unloaded!!!
vlad@DESKTOP-MKU600H:/dev$

```

Продолжаем наше задание. Теперь создаем `test3.c` для нашего 3 и 4 задания.

```
$ make
```

```
$ sudo insmod test3.ko
```

```

vlad@DESKTOP-MKU600H:~/eltex/hw_20$ sudo insmod test3.ko
[sudo] password for vlad:
vlad@DESKTOP-MKU600H:~/eltex/hw_20$ dmesg | tail -n10
[ 859.634420] mini_init (168): drop_caches: 1
[ 955.487541] mini_init (168): drop_caches: 1
[1435.656643] mini_init (168): drop_caches: 1
[1532.817255] mini_init (168): drop_caches: 1
[1951.650779] mini_init (168): drop_caches: 1
[2108.565582] mini_init (168): drop_caches: 1
[2172.613314] mini_init (168): drop_caches: 1
[2318.767568] test3: loading out-of-tree module taints kernel.
[2318.771567] My test module loaded!!!!
[2318.771573] Major number is 239
vlad@DESKTOP-MKU600H:~/eltex/hw_20$

```

Проверяем создалась ли директория в **/proc**:

mc [vlad@DESKTOP-MKU600H: ~] × + ▾					
Left	File	Command	Options	Right	
< /proc				.[^]>	
.n	Name		Size	Modify	time
locks			0	Jun 16	16:28
meminfo			0	Jun 16	16:27
misc			0	Jun 16	16:28
modules			0	Jun 16	16:28
@mounts			11	Jun 16	16:27
mtrr			0	Jun 16	16:28
pagetypeinfo			0	Jun 16	16:28
partitions			0	Jun 16	16:28
schedstat			0	Jun 16	16:28
slabinfo			0	Jun 16	16:28
softirqs			0	Jun 16	16:28
stat			0	Jun 16	16:28
swaps			0	Jun 16	16:27
sysrq-trigger			0	Jun 16	16:28
test			0	Jun 16	17:05
timer_list			0	Jun 16	16:28
uptime			0	Jun 16	16:27
version			0	Jun 16	16:28
vmallocinfo			0	Jun 16	16:28
vmstat			0	Jun 16	16:28
zoneinfo			0	Jun 16	16:28
test					


```

vlad@DESKTOP-MKU600H:/proc$ cat test
Hello
vlad@DESKTOP-MKU600H:/proc$ |
vlad@DESKTOP-MKU600H:/proc$ echo "ASDASDZX" > test
vlad@DESKTOP-MKU600H:/proc$ cat test
ASDASDZX
vlad@DESKTOP-MKU600H:/proc$ |

```

а также объект ядра в **/sys**:

mc [vlad@DESKTOP-MKU600H]
+
v

Left	File	Command	Options	Right
<-	/sys/kernel/kobject_test			.[^]>
.n	Name		Size	Modify time
/..			UP--DIR	Jun 16 16:27
test_string			4096	Jun 16 17:08

UP--DIR

Hint: Shell commands will not work when you are on a non-local

```

vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ |
vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ cat test_string
ASDASDZX
vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ |

```

И по старой схеме выгружаем то, что загрузили:

```
vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ sudo rmmod test3
vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ dmesg | tail -n10
[ 955.487541] mini_init (168): drop_caches: 1
[ 1435.656643] mini_init (168): drop_caches: 1
[ 1532.817255] mini_init (168): drop_caches: 1
[ 1951.650779] mini_init (168): drop_caches: 1
[ 2108.565582] mini_init (168): drop_caches: 1
[ 2172.613314] mini_init (168): drop_caches: 1
[ 2318.767568] test3: loading out-of-tree module taints kernel.
[ 2318.771567] My test module loaded!!!!
[ 2318.771573] Major number is 239
[ 2536.345694] My test module unloaded!!!
vlad@DESKTOP-MKU600H:/sys/kernel/kobject_test$ |
```